



Incident report analysis

Project Overview

This project involves analyzing a real-world cybersecurity incident — a DoS attack that disrupted a multimedia company's internal network for two hours — and using it to build a comprehensive security improvement plan. Following the NIST Cybersecurity Framework (CSF), the task is to walk through five core functions: Identify, Protect, Detect, Respond, and Recover.

The goal is to examine how the attack happened, what vulnerabilities were exploited, and what the organization can do differently going forward. Each section of the framework guides the analysis — from auditing network risks and strengthening defenses, to improving monitoring, establishing a response plan, and ensuring faster recovery in future incidents. The end result is a structured, actionable security strategy tailored to the organization's needs.

Summary	Security Event Summary The company experienced a DoS attack where a malicious actor flooded the network with ICMP packets, causing network services to stop responding for about two hours . The attack was possible due to an unconfigured firewall . The incident response team blocked ICMP traffic, shut down non-critical services, and restored critical systems. Afterwards, the security team implemented ICMP rate-limiting, IP verification, network monitoring, and
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	an IDS/IPS to prevent similar attacks.
Identify	The attack was an ICMP flood DoS attack, where a malicious actor exploited an unconfigured firewall to overwhelm the network with ping packets. The affected systems included the company's entire internal network infrastructure, rendering all network resources inaccessible to legitimate users. No specific data breach was identified, but the disruption impacted all internal services for approximately two hours.
Protect	The organization should immediately patch the misconfigured firewall and enforce strict ICMP rate-limiting rules across all network entry points. Access control policies should be reviewed and updated to ensure no firewall rules are left unconfigured or unmonitored. Additionally, regular firewall audits and employee security awareness training should be implemented to prevent similar vulnerabilities from being exploited in the future.
Detect	The organization should deploy network monitoring software and an IDS/IPS system to continuously analyze traffic patterns and flag anomalous activity, such as unexpected ICMP floods or unrecognized source IPs. Firewall logs should be reviewed regularly to track authorized versus unauthorized access attempts, with automated alerts configured for suspicious behavior. Additionally, user account activity should be monitored through a SIEM tool to detect unusual login patterns or unauthorized access in real time.

Respond	In a future incident, the team should immediately isolate affected devices and block suspicious traffic at the firewall to contain the threat. IDS/IPS alerts, firewall logs, and traffic data should be used to neutralize and analyze the attack. Recovery can be improved by maintaining tested backups, defining clear team roles, and setting recovery time objectives to restore services faster than the two hours experienced in this event.
Recover	To recover immediately, the team needs access to system backups, firewall configuration records, and a list of critical versus non-critical services to prioritize restoration. Recovery processes include restoring critical network services first, applying updated firewall rules, and validating system integrity before bringing all services back online. A post-incident review should then be conducted to document lessons learned and refine the recovery plan for future incidents.

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